These points divides the whole interval into three intervals namely $(-\infty,-2),(-2,3)$ and $(3,\infty)$.



Fig : 6.11

Interval	Sign of $f'(x)$	Intervals of increasing/ decreasing
$(-\infty, -2)$	$(-)(-) > 0$	Increasing in $(-\infty, -2]$
$(-2, 3)$	$(-)(+) < 0$	Decreasing in $[-2, 3]$
$(3, \infty)$	$(+)(+) > 0$	Increasing in $[3, \infty)$

Table: 6.2

Example 6.21

Find the stationary value and the stationary points $f(x)=x^2+2x-5$.

Solution:

Given that $f(x)= x^2 + 2x - 5$ … (1)

$f'(x) = 2x + 2$

At stationary points, $f'(x) = 0$

$\Rightarrow 2x + 2 = 0 \Rightarrow x = -1$

$f(x)$ has stationary value at $x = -1$

When $x = -1$, from (1)

$f(-1) = (-1)^2+2(-1)-5 = -6$

Stationary value of $f(x)$ is -6

Hence stationary point is $(-1,-6)$

Example 6.22

Find the stationary values and stationary points for the function: $f(x)=2x^3+9x^2+12x+1$.

Solution :

Given that $f(x) = 2x^3+9x^2+12x+1$.

$f'(x) = 6x^2+18x+12$

$= 6(x^2+3x+2) = 6(x+2)(x+1)$

$f'(x) = 0 \Rightarrow 6(x+2)(x+1) = 0$

$\Rightarrow x + 2 = 0$ (or) $x + 1 = 0$.

$\Rightarrow x = -2$ (or) $x = -1$

$f(x)$ has stationary points at $x = -2$ and $x = -1$

Stationary values are obtained by putting $x = -2$ and $x = -1$.

When $x = -2, f(-2)=2(-8)+9(4)+12(-2)+1= -3$

When $x= -1, f(-1)= 2(-1)+9(1)+12(-1)+1=-4$

The stationary points are $(-2,-3)$ and $(-1,-4)$.

Example 6.23

The profit function of a firm in producing x units of a product is given by $P(x)=\frac{x^3}{3}+x^2+x$. Check whether the firm is running a profitable business or not.

Solution:

$P(x) = \frac{x^3}{3}+x^2+x.$

$P'(x) = x^2+2x+1=(x+1)^2$

It is clear that $P'(x)>0$ for all x.

$\therefore$ The firm is running a profitable business.

IMPORTANT NOTE

Let $R(x)$ and $C(x)$ are revenue function and cost function respectively when x units of commodity is produced. If $R(x)$ and $C(x)$ are differentiable for all $x > 0$, then $P(x) = R(x) - C(x)$ is maximized when Marginal Revenue = Marginal cost. i.e. when $R'(x)=C'(x)$ profit is maximum at its stationary point.

Example 6.24

Given $C(x)=\frac{x^2}{6}+5x+200$ and $p(x) = 40-x$ are the cost price and selling price when x units of commodity are produced. Find the level of the production that maximize the profit.

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